Elements of Mathematical Linguistics

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1 Introduction

It is no coincidence that many linguistic researchers have a background in mathematics. As the scientific study of language, linguistics requires well-defined mathematical models, and the mathematical precision of language is indispensable in testing and implementing theories of language. Without the helping hand of mathematics, the crucial questions of the extensive grammars in computational applications of language processing cannot be addressed. Since language is structured at several levels (phonology, morphology, syntax, and semantics), mathematical concepts and techniques have been developed and applied to each area, thus forming subdivisions in mathematical linguistics.

2 Probability Meets Poetry

In January 1913, Andrei Markov perused the first 20,000 words of Pushkin's Eugene Onegin, painstakingly counting vowels (8,638) and consonants (11,362) and noting the order of their appearance as vowel-vowel pairs (1,104), consonant-consonant pairs (3,827) and combinations of the two speech sounds (15,069). Interestingly enough, his analysis started with the goal of rebuffing another mathematician, Pavel Nekrasov, with whom he had disputed for years. Specifically, Nekrasov's claim (rooted in his theological belief of free will) that the Weak Law of Large Numbers (the idea that as a sample size becomes larger, the sample mean gets closer to the expected value) applied only to independent variables, set Markov on the inquiry to prove Nekrasov's assertion incorrect. At the end, Markov would show that the Law of Large Numbers applies to systems of dependent variables.

Markov divided the whole 20,000 letter sequence into 200 separate sequences of 100 letters, which were arranged into 10 by 10 tables. He then counted the vowels in each column of the table, and added the numbers of vowels from the first and sixth columns, second and seventh columns, third and eighth columns, etc. (Those sums are written vertically as seen below.) Finally, he combined each vertically written number. (We see the final sums as written in bold.) The most frequent value of vowels among the listed sums was 43 (as seen below). Markov calculated the mean number of vowels per block of 100 letters to be approximately 43, and the variance to be 2.26. Independence of chunks of letters (one 100-letter sequence had no effect on the next 100-letter sequence) and randomness of the sample followed a normal distribution.

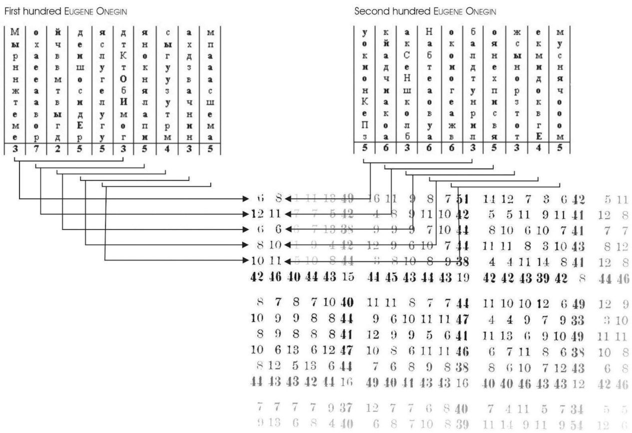

Figure 1: Markov's counting method

Next, he turned to calculate the probability of two vowels in succession. Assuming two adjacent letters were independent, the probability would be equal to 0.43 * 0.43, or about 0.19. In such a case, our sample of letters would be expected to have about 3,800 vowel pairs out of the 20,000 letters. This count, however, exceeded the actual number of the vowel-vowel pairings appearing in the text (1,104) by threefold. In addition, given phonetic features of language, there exists a strong tendency for alternating vowels and consonants. Hence, there was an evident dependency among letters; in a sequence or "chain" of letters, each letter was dependent on the preceding one. For example, a certain batch of consonants forced the next letter to be a vowel. Such events, linking one to the next, formed what came to be known as a Markov chain, in which the future state of the system depended only on the current state. Markov chains can be represented as probabilistic state diagrams where the transitions between events or states are described with the probabilities of their occurrences.

3 Shannon and Language Modeling

Markov's investigation into probabilistic letter sequences found its way into Claude Shannon's 1948 paper, "A Mathematical Theory of Communication," and was used as a mathematical model to predict the sequences of words in a piece of English text. First, Shannon used a Markov model of order 0 to make a prediction of how likely a given letter occurs in the texts; in other words, based on its frequency, he would predict a fixed probability of the letter. For example, in the text "abcbbaccc," the predictability of 'a' occurring is 2/9, 'b' 3/9 and 'c' 4/9. According to this model, based on independence assumption and given probability, any stream of letters could be effectively generated. However, as groups of letters in the English language occur together with certain degrees of correlation ('t' is more often followed by 'h' and not 'x'), Shannon expanded this Markov model by relating the probability of a chosen letter to its successive or preceding letters. Then, Shannon explored the idea of higher-order Markov models: bigrams (pairs of words), trigrams (three successive words), and n-grams (a sequence of n words) with each state dependent on 2, 3 or n prior steps.

He thought that higher order Markov models would get us closer to understanding the fundamental concepts of language. However, with the rise of the Chomskyan approach to human grammatical knowledge and the argument that Markov chains did not capture relationships between words, research into application of probability theory to linguistics stagnated for decades.

Before discussing the grammatical approach to linguistics, let's turn to the notion of sets in the domain of language.

4 Notation

A mathematical concept such as set theory is important in language study because language can be viewed, on a basic level, as a collection of elements such as strings or words which, when stitched together, form sentences. Hence, just as sets, sentence structures are abstract constructs containing elements that follow certain descriptions. Let's start with strings, which are fundamental structures of language. (We cannot read this article without understanding their workings.) An alphabet, denoted as "Σ," can be defined as a finite set of symbols. Any string created from an alphabet is a finite set of symbols of Σ. Any set of such strings is referred to as a formal language over an alphabet. For example, in an alphabet Σ with symbols a, b, c, a certain language must have strings that start with symbol b and contain two characters. This means that the possible strings for this language include "ba," "bb," and "bc."

A language can be further characterized as "Σn" (the set of strings with a length of n), "$\Sigma+$" (the set of strings with a length of 1 or greater, also known as the positive closure of Σ), and "$\Sigma*$" (the set of all strings, also known as the Kleene closure or Σ).

5 Chomsky and "colorless green ideas sleep furiously"

In the 1950s, in an attempt to understand the principles that govern human speech, linguist Noam Chomsky defined a set of rules called grammars, which specified the structural description of each sentence. The sentence "colorless green ideas sleep furiously" follows the rules of syntax, yet it is meaningless; grouping words in a linear order does not sufficiently capture the unit's meaning. It illustrates Chomsky's point that the probabilistic approach to syntax takes us only halfway to the goal of generating language to convey meaning.

Using grammar structures such as

$$\begin{gathered}\text{DNP VP} \rightarrow \text{S} \\ \text{V DNP} \rightarrow \text{VP} \\ \text{P DNP} \rightarrow \text{PP},\end{gathered}$$

where

- DNP = definite noun phrase

- VP = verb phrase
- S = sentence
- V = verb
- P = preposition
- PP = prepositional phrase

and a parse tree, Chomsky broke down human speech patterns as each word corresponded to a given type.

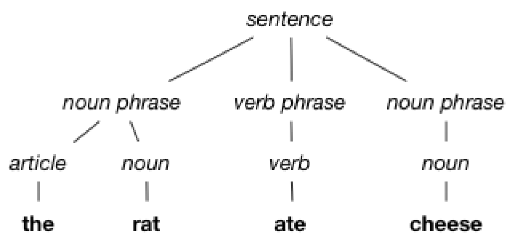

Figure 2: Parse tree of a simple sentence

This hierarchical structure of language is used extensively in natural language processing (NLP), which is the concept of artificial comprehension of the human language. The process of breaking down sentences into their components, known as tokenization, allows for machines to understand (or not understand, in the case of a nonsensical phrase) written language.

Another such implementation of language hierarchy is in the finite-state automata, which is a machine that monitors and changes state based on inputs. Upon processing each input, the machine adjusts for the most likely successive input. For example, a finite-state machine can process alphabetical inputs and predict phrases by utilizing logical grammar structures based on parse trees.

6 Conclusion

Overall, the goal of mathematical linguistics is to gain a better understanding of the nature of human language and how it is processed by the brain. Over the decades, mathematical methods to analyze multiple structural levels of language have been developed, changed and adopted. Markov's probabilistic sequential text analysis proved to be a momentous, early example of the application of mathematical techniques to the study of language, and it paved the way for further research in the field. Incorporating Markov's models, Shannon's work gave a greater insight into the relationship between successive vowels and consonants in forming words and sentences. (A common application of Markov chains is text prediction; when users write emails, Google predicts words to autocomplete it or suggests replies to received messages.) In the 1950s, there was a gradual shift from statistical modeling to theoretical foundations where attention was paid to formal logical approaches to language. However, in the 1970s, enabled by increasing availability of machine-readable text data, the probability theory in linguistics reemerged, and since then has produced more sophisticated probabilistic models. Currently, the corpus-based statistical training and parsing techniques are used in conjunction with modern language modeling done with neural network language models. This development in finding and using mathematical models to solve linguistic problems will continue, thanks to larger and more accessible linguistic data, increased research in AI, and the processing power of modern machines.

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